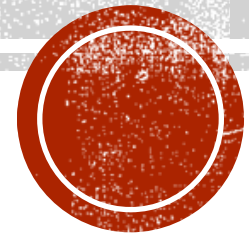


LECTURE 05

Selection Widgets



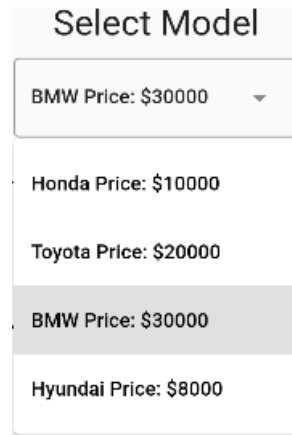
FLUTTER SELECTION WIDGETS

- ❑ Flutter has many selection widgets, that allows the user to choose from a set of predefined values.
- ❑ We will discuss 3 basic selection widgets in this presentation:
 - The DropDownMenu widget
 - The Radio button widget
 - Checkbox widget
- ❑ Selection widgets forces the user to select predefined values and minimizes application errors at runtime.



DROPDOWNMENU WIDGETS

- ❑ DropDownMenu widget is a collapsible widget that displays a menu when you press on it.
- ❑ It saves screen space by displaying the options when pressed only.
- ❑ You populate the widget using a list. The list can contain any type of objects.



RADIO WIDGETS

- ❑ Radio widgets allows the user to select among multiple options.
- ❑ The user can select only one Radio button from each group of radio buttons.
- ❑ Radio buttons take more screen space but is more convenient for the user. It is used when we have a limited number of options.

Warranty ☒ 1 year ☐ 5 years



CHECKBOX WIDGETS

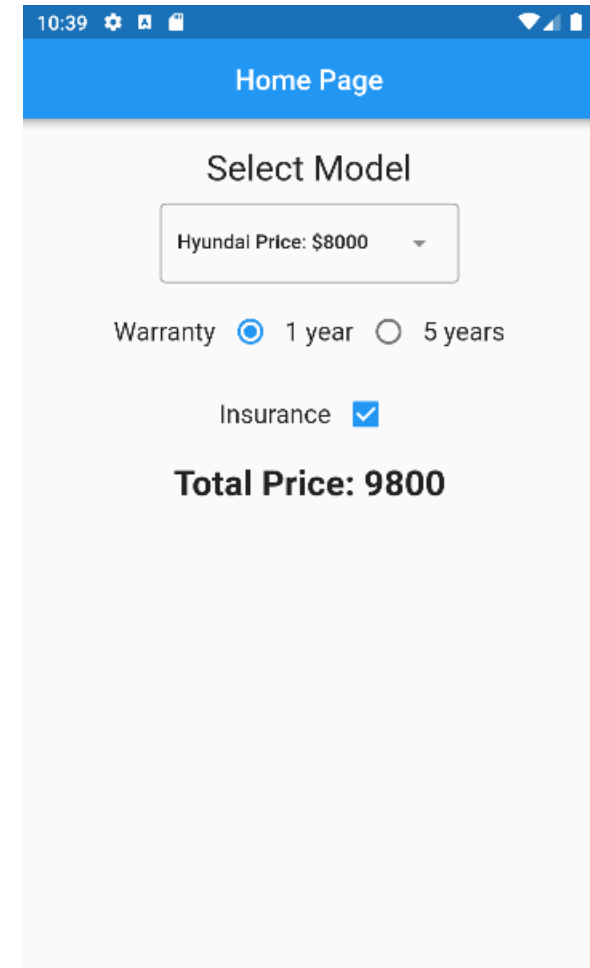
- ❑ Checkbox widgets allows the user to select/deselect a specific option.
- ❑ They use less space than Radio widgets.
- ❑ Below is an example where the user can choose to add car insurance.

Insurance ☒



THE APPLICATION

- ❑ We will create an application that contains the 3 types of selection widgets.
- ❑ The user will be asked to select car model, years of warranty and insurance option. The application will then calculate the total car price.



The screenshot shows a mobile application interface with a blue header bar labeled "Home Page". Below the header, the title "Select Model" is centered. A dropdown menu is displayed with the text "Hyundai Price: \$8000" and a downward arrow. Below the dropdown, there are two radio button options for "Warranty": "1 year" (selected) and "5 years". Below the warranty options, there is a checkbox for "Insurance" which is checked. At the bottom, the text "Total Price: 9800" is displayed in bold. The status bar at the top shows the time "10:39" and various icons.



CREATING THE CAR CLASS

We will create a Car class and place it in the car.dart file. The code is below:

```
class Car {  
  cl  
  String model;  
  int price;  
  int warranty = 1; // warranty years, default 1 year  
  bool insurance = false; // determines if car is insured, default is false  
  // constructor only takes model and price. warranty and insurance are set later  
  Car(this.model, this.price);  
  // toString method used to display an item in a dropdown widget  
  @override  
  String toString() {  
    return '$model Price: \$$price';  
  }  
}
```

```
String getTotalPrice() {  
  /*  
    calculate price as follows: if warranty is 1 then 5% else 10%  
    if insurance is added, then add 1000 to total price  
  */  
  int insuranceAmount = insurance ? 1000 : 0;  
  if (warranty == 1) {  
    return (price * 1.05 + insuranceAmount).toStringAsFixed(0);  
  }  
  return (price * 1.1 + insuranceAmount).toStringAsFixed(0);  
}
```

CREATING THE CARS LIST

We will create a list of cars list to populate the DropdownMenu later.

```
// a constant list of car objects. It is used to populate the items of  
// dropdown widget
```

```
List<Car> cars = [  
    Car('Honda', 10000),  
    Car('Toyota', 20000),  
    Car('BMW', 30000),  
    Car('Hyundai', 8000)  
];
```



CREATING THE MYDROPDOWNMENUWIDGET

Create a new stateful widget in the car.dart file. The widget class contains the below code:

```
class MyDropdownMenuWidget extends StatefulWidget {  
  const MyDropdownMenuWidget({required this.updateCar, super.key});  
  // below is a callback function to return the selected car to the home page  
  // we will call it from the widget's state class  
  final Function(Car) updateCar;  
  
  @override  
  State<MyDropdownMenuWidget> createState() => _MyDropdownMenuWidgetState();  
}
```



MYDROPDOWNMENUWIDGET STATE CLASS

```
class _MyDropdownMenuWidgetState extends State<MyDropdownMenuWidget> {  
  /*  
  class to represent a dropdown of cars.  
  It is populated using the cars list.  
  It contains a callback function called updateCar to pass the selected car back  
  to the home page.  
  */  
  @override  
  Widget build(BuildContext context) {  
    return DropdownMenu(  
      width: 210.0,  
      initialSelection: cars[0], // first car to be displayed  
      onSelected: (car) {  
        setState(() {  
          widget.updateCar(car as Car); // use widget to access widget fields from state class  
        });  
      }, // the list map function converts the list of cars to a list of DropdownMenuEntries  
      dropdownMenuEntries: cars.map<DropdownMenuEntry<Car>>((Car car) {  
        return DropdownMenuEntry(value: car, label: car.toString());  
      }).toList());  
  }  
}
```

CREATING THE WARRANTYWIDGET

Create a new stateful widget in the car.dart file. The widget class contains the below code:

```
class WarrantyWidget extends StatefulWidget {  
  /*  
    class to represent radio button group for selecting warranty years  
    The default is one year  
    It includes a callback method called updateWarranty, to return the years to  
    home page.  
  */  
  WarrantyWidget({required this.car, required this.updateWarranty, Key? key}) : super(key:  
key);  
  final Function(int) updateWarranty;  
  Car car;  
  
  @override  
  State<WarrantyWidget> createState() => WarrantyWidgetState();  
}
```



WARRANTY WIDGET STATE CLASS

```
class WarrantyWidgetState extends State<WarrantyWidget> {  
  int _years = 1; // stores warranty years and is used so that both radio buttons belong to the same group  
  
  @override  
  Widget build(BuildContext context) {  
    // we add the radio group to a row widget, to arrange them horizontally  
    return Row(mainAxisAlignment: MainAxisAlignment.center, children: [  
      const Text('Warranty', style: TextStyle(fontSize: 18.0)),  
      Radio( value: 1, groupValue: widget.car.warranty, // all radio buttons in the same group should share a common variable  
        onChanged: (val) {  
          setState(() {  
            _years = val as int;  
            widget.updateWarranty(_years); // use widget to access widget fields from state class });  
          },  
        ), const Text('1 year', style: TextStyle(fontSize: 18.0)),  
      Radio( value: 5, groupValue: widget.car.warranty, // all radio buttons in the same group should share a common variable  
        onChanged: (val) {  
          setState(() {  
            _years = val as int;  
            widget.updateWarranty(_years); // use widget to access widget fields from state class });  
          },  
        ), const Text('5 years', style: TextStyle(fontSize: 18.0)),  
    ]);  
  }  
}
```

THE HOME.DART FILE

```
/*  
Page to display the main interface of the application.  
it displays the all widgets horizontally.  
It allows the user to select the car model and price form a dropdown.  
The user selects the warranty years form a radio group  
The user can add car insurance via checkbox  
The total price is automatically calculated when one of the widget's state changes  
*/  
  
import 'package:flutter/material.dart';  
import 'car.dart';  
  
class Home extends StatefulWidget {  
  const Home({super.key});  
  
  @override  
  State<Home> createState() => _HomeState();  
}
```



HOMEWIDGET STATE CLASS

```
class _HomeState extends State<Home> {  
  String totalPrice = cars.first.getTotalPrice(); // holds total car price  
  Car car = cars.first; // set the first car to be displayed  
  
  void updateCar(Car car) {  
    // updates car price when the user selects a car form the dropdown  
    setState(() {  
      this.car = car;  
      totalPrice = car.getTotalPrice();  
    });  
  }  
  
  void updateWarranty(int warranty) {  
    // updates warranty years when the user clicks on a radio buttons  
    setState(() {  
      car.warranty = warranty;  
      totalPrice = car.getTotalPrice();  
    });  
  }  
}
```



HOMEWIDGET STATE CLASS BUILD METHOD

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Center(
      child: Column(children: [
        const SizedBox(height: 20.0),
        const Text('Select Model', style: TextStyle(fontSize: 25.0)), const SizedBox(height: 10.0),
        MyDropDownMenuWidget(updateCar: updateCar), const SizedBox(height: 10.0),
        WarrantyWidget(updateWarranty: updateWarranty, car: car), const SizedBox(height: 10.0),
        Row(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            const Text('Insurance', style: TextStyle(fontSize: 18.0)),
            Checkbox(value: car.insurance, onChanged: (bool? value) {
              setState(() { // updates the total price when the checkbox state changes
                car.insurance = value as bool;
                totalPrice = car.getTotalPrice();
              });
            })
          ]
        ),
        const SizedBox(height: 10.0),
        Text('Total Price: $totalPrice', style: const TextStyle(fontSize: 25.0, fontWeight: FontWeight.bold)),
      ]
    )
  );
}
```

THE MAIN.DART FILE

```
import 'package:flutter/material.dart';
import 'home.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({Key? key}) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'csci410 week 6',
      home: Home() // calls the main page to display the application interface
    );
  }
}
```

